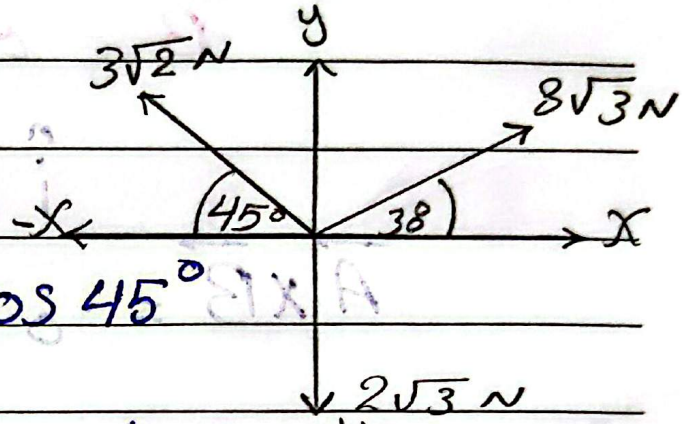
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section 2

Find the resultant of the forces which appear in the following diagram.



$$x = 8\sqrt{3}\cos 30^\circ - 3\sqrt{2}\cos 45^\circ$$

$$= 8\sqrt{3} \times \frac{\sqrt{3}}{2} - 3\sqrt{2} \times \frac{1}{\sqrt{2}}$$

$$\boxed{x = 12 - 3 = 9\ N}$$

$$y = 8\sqrt{3}\sin 30 + 3\sqrt{2}\sin 45^\circ - 2\sqrt{3}$$

$$= 8\sqrt{3} \times \frac{1}{2} + 3\sqrt{2} \times \frac{1}{\sqrt{2}} - 2\sqrt{3}$$

$$\boxed{y = (2\sqrt{3} + 3)\ N}$$

$$R = \sqrt{x^2 + y^2} = \sqrt{9^2 + (2\sqrt{3} + 3)^2} = 11.1\ N$$

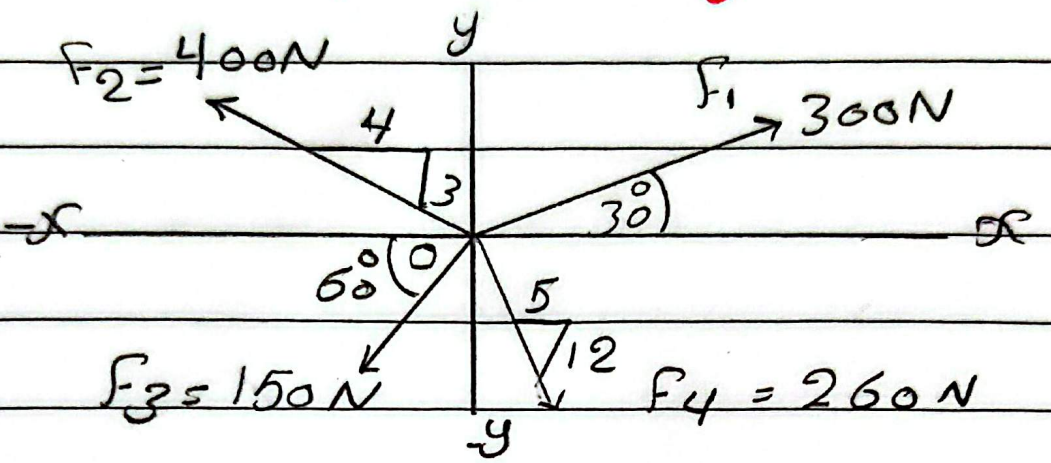
$$\tan\theta = \frac{y}{x} = \frac{2\sqrt{3} + 3}{9} \Rightarrow \therefore \theta = 35,69^\circ$$

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Determine the resultant of four forces acting an abody as shown below



$R^x = 259.8 - 320 - 75 + 100 = -35.2N$

$R^y = 150 + 240 - 130 - 240 = 20N$

$R = \sqrt{(-35.2)^2 + (20)^2}$

$\tan\theta = \frac{R^y}{R^x} = \frac{20}{35.2}$

$\theta = 29.6^\circ$

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Find the moment of the force 20 N which effect on the straight line $\overrightarrow{AB}$, where A(5, 3, -3), B(2, 2, 1) about the origin.

$$\hat{e} = \frac{\overrightarrow{AB}}{|\overrightarrow{AB}|} = \frac{\overrightarrow{B-A}}{|\overrightarrow{AB}|} = \frac{(2-5)\hat{i}+(2-3)\hat{j}+(-1+3)\hat{k}}{\sqrt{(2-5)^2+(2-3)^2+(-1+3)^2}}$$

$$= \frac{1}{\sqrt{14}}(-3\hat{i}-\hat{j}+2\hat{k})$$

$$\overrightarrow{F} = \frac{20}{\sqrt{14}}(-3\hat{i}-\hat{j}+2\hat{k})$$

$$\overrightarrow{r} = 2\hat{i}+2\hat{j}-\hat{k}$$

$$M_O = \overrightarrow{r} \times \overrightarrow{F} = \frac{1}{\sqrt{14}} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & -1 \\ -3 & -1 & 2 \end{vmatrix}$$

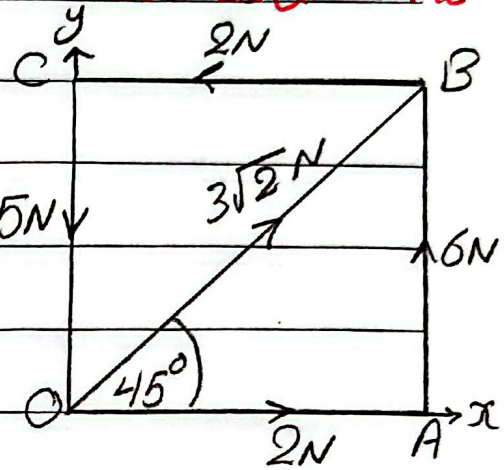
$$= \frac{1}{\sqrt{14}}(3\hat{i}-\hat{j}+4\hat{k})$$

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if the side for a square OABC is 1m the forces 2, 6, 2, 5, $3\sqrt{2}$ affected the sides OA, AB, BC, Co, OB respectively. Find the magnitude, the direction, and the line of the resultant forces.



$$R_x = 2 + 3\sqrt{2}\cos 45^\circ - 2$$

$$= (3\sqrt{2})\left(\frac{1}{\sqrt{2}}\right) = 3N$$

$$R_y = 6 + 3\sqrt{2}\sin 45^\circ - 5$$

$$= 1 + (3\sqrt{2})\left(\frac{1}{\sqrt{2}}\right) = 4N$$

$$M_o = (6)(1) + 2(1) = 8\,Nm$$

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{3^2 + 4^2} = 5N$$

$$\theta = \tan^{-1}\frac{R_y}{R_x} = \tan^{-1}\frac{4}{3} = 53,13^\circ$$

$$M_o - xR_y + yR_x = 0$$

$$8 - 4x + 3y = 0 \quad \text{or} \quad 4x - 3y + 8 = 0$$

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